

Quarterly - Based Git Branching Strategy

In our project, we follow a quarterly release model. That means,

- Every 3 months, we have one main release,
- For each quarter, we create a separate main release branch

Example :-

- For Quarter 1 (Jan to Mar) → Branch name 2024 Q1
- For Quarter 2 (Apr to Jun) → Branch name 2024 Q2
- This branch is created from 2024 Q1

Types of Branches

① Main Release Branch

- This is the main branch for the quarter, like 2024 Q1, 2024 Q2, etc.
- All the final features and bug fixes are merged here after testing.
- Once the quarter ends, this branch is used to deploy the final code to production.

② Feature Branch

- When developers start working on a new feature, they create a feature branch.
- This branch is usually created from the current main release branch.

Example:

```
git checkout -b feature-login-2024Q2
```

- After Development and testing, the feature branch is merged into the main release branch.

Back merge (optional)

- If the same feature is also needed in previous releases (e.g., 2024 Q1), we do a back merge.

③ Bug Fix Branch

- If a client finds a bug in a specific release (e.g., Q1) we create a bug fix branch from that main branch.

Example:

```
git checkout -b bugfix-123-2024Q1
```


once the Bug is fixed and tested, it is delivered to the client

Forward merge:

if the same bug is found in later release (eg. 2024 Q2), we Forward merge the fix.

```
git checkout 2024Q2
git merge bugfix-login-issue
```

Backward merge:

If the fix is needed in older stable-active releases, we back merge it there too.

visual Branching flow

main Branches: 2024 Q1 → 2024 Q2 → 2024 Q3

Feature Branch: ↘
→ Feature - login (from 2024 Q2)

Back merge to 2024 Q1 ← merge into Q2 → Forward merge to 2024 Q3

Bug Fix Branch:

from 2024 Q1 → bugfix - login - issue

merge back into Q1

merge forward to Q2

Summary Table:

<u>Branch type</u>	<u>Created from</u>	<u>Purpose</u>
main Release	previous quarter	Quarterly production release
Feature Branch	main Release	develop and test new features.

Bug Fix Branch	main Release	Fix issues reported by client
Backward Merge	Later branch	copy features or fix to older branch
forward merge	Earlier branch	copy bug fix to future branches.